Robert Ondino Morelli

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Education

B.S. Computer Science | *University of Utah* | *Expected May 2027*

Skills

Languages

- Zig, C/C++, Python, Java, TypeScript/JS

Tools

- Git, Docker, GitHub Actions, CI/CD

Experience

Research Assistant | *University of Utah* | *October 2025 – Present*

- Analyzing performance cost of gradual typing in Meta's Static Python (CinderX)
- Automatic rewriting of Python code to construct test examples
- First author, "Automatic Cast Insertion to Evaluate Progressive Gradual Types" (under review)

Teaching Assistant | *University of Utah* | *September 2025 – May 2026*

- Led labs, graded, and assisted students for COMP 1020

Software Engineer/DevOps | *Stutor Inc.* | *September 2023 – April 2024*

- Architected automation pipeline using Fastlane and GitHub Actions to build, sign and deploy to multiple platforms
- Added compound keys to tables where backend queries had drifted from database schema; query times reduced from 15s+ to <1s

Web Developer/DevOps | *Jennan Software Solutions* | *April 2022 – September 2023*

- Software consulting. Clients included LDS Church, Retro Game Vault and Stutor Inc.
- Architected, proposed and implemented overhaul of development process for the LDS MTC team using build automation in conjunction with new QA practices. Deployment was reduced from a week-long headache to mere minutes
- Rewrote Embark startup using dependency-ordered queue for async initializations; First-time loading reduced by 50%

Research Assistant Intern | *Earl Keefe PhD, University of Utah* | *November 2020 – June 2021*

- Used pypng to construct novel visualizations of covariance matrices to provide more intuitive understanding of data
- Implemented statistical formulas in clear code in order that a reviewer can be sure the results are correct and valid

Web Development Intern | *Frelii* | *June 2019 – September 2019*

- Implemented web scraping SNPedia for AI training using python with multiprocessing and BeautifulSoup

Projects

IRX regex fuzzer | *Zig* | ↗

- The inverted regex fuzzer generates accept and reject string seeds and fuzzes patterns instead of inputs across ~25 engines

UTU language and compiler | *JS/TS, WasmGC* | ↗

- The language AssemblyScript was meant to be. A language designed from the ground up to have predictable lowering to WasmGC, have small bundle size, and support .d.ts as an ABI

Spreadsheet formulas to DLL compiler | *C#* | ↗

- Compiles spreadsheet formulas into a DLL, supporting optimizations including fast-math style unsafe float conversion

Color alchemy | *Qt/C++* | ↗

- Beautiful game for learning color mixing, properly using OKLab color space for gradient construction and scoring

Misc

Contributed to open-source IDE code-base | *MARS* | ↗

- Discovered and corrected oversight in the MARS IDE's instruction decoder that affected correctness of self-modifying code

CinderX bug hunting | *CinderX* |

- Reported 11 bugs in CinderX and contributed 2 fixes to the type-checker

Regex bug hunting

- Reported 46 correctness bugs across a wide variety of regular expression engines including Perl, .NET and V8

Competitive Programming | *ICPC* | ↗

- As part of team "Big Cottonwood Coders", placed 12 out of 48 teams at the Rocky Mountain regional contest (2025)
- As Event Coordinator of the U of U technical coding club, organized and oversaw competitions (2026)

Jane Street Leaderboard

- **Number Cross 5** ↗
- **Sum One, Somewhere** ↗